AP CALCULUS AB

Unit 1 • Lesson 1.6 practice

Lesson 1.6 • Determining Limits Using Algebraic Manipulations

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to -2} \dfrac{x^2 - 4}{x + 2}$ is

(A) 0

(B) -4

(C) 4

(D) It does not exist.

Question 2

NO CALCULATOR

$\lim_{x \to 4} \dfrac{\sqrt{x} - 2}{x - 4}$ is

(A) $\dfrac{1}{4}$

(B) $\dfrac{1}{2}$

(C) 0

(D) 4

Question 3

NO CALCULATOR

$\lim_{h \to 0} \dfrac{(3 + h)^2 - 9}{h}$ is

(A) 9

(B) 0

(C) 6

(D) 3

Question 4

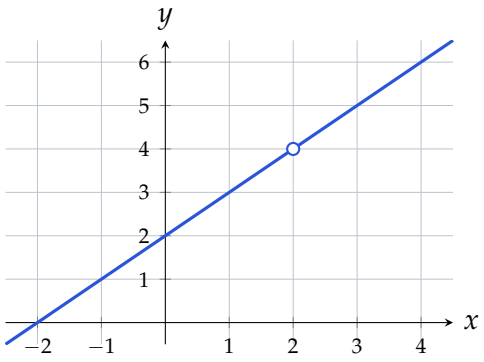
NO CALCULATOR

$\lim_{x \to 1} \dfrac{\frac{1}{x} - 1}{x - 1}$ is

(A) 1
(B) 0
(C) It does not exist.
(D) -1

Question 5 (graph) NO CALCULATOR

The graph of $f(x) = \dfrac{x^2-4}{x-2}$ is shown. What is $\lim\limits_{x\to 2} f(x)$?



(A) 0
(B) 4
(C) 2
(D) It does not exist.

Question 6 (Free response) NO CALCULATOR

Evaluate each limit. Show the algebra.

(a) $\lim\limits_{x\to 5} \dfrac{x^2-3x-10}{x-5}$ (b) $\lim\limits_{x\to 0} \dfrac{\sqrt{4+x}-2}{x}$

(c) $\lim\limits_{x\to 2} \dfrac{x^3-8}{x-2}$ (d) $\lim\limits_{h\to 0} \dfrac{\frac{1}{2+h}-\frac{1}{2}}{h}$